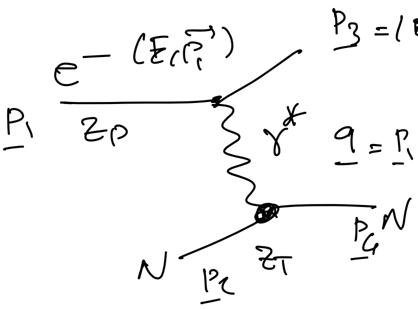


$P_3 = (E', \vec{p}_3{}')$ Spin ½ prob e^- $e^- + N \to e^- + N$

$q = P_1 - P_3$ Spinless target with charge extension / structure

$\rho(r) = Z_T e \, f(\vec{r})$

$q^2 \equiv -Q^2$

$Q^2 \equiv -q^2 = 4EE' \sin^2\frac{\theta}{2}$ $\nu = E - E'$

$$\frac{d\sigma}{d\Omega} = \frac{(\alpha Z_T Z_P)^2}{q^4} E'^2 \left(1 - \beta^2 \sin^2\frac{\theta}{2}\right) \frac{E}{E'} \; |F(q^2)|^2$$

$$F(q^2) = (\text{Norm}) \int d^3r \; e^{+i\vec{q}\cdot\vec{r}} \; f(\vec{r})$$ Form Factor

$$\boxed{\text{TOMORROW TUESDAY @ 8:30}}$$

$$\frac{\left.\frac{d\sigma}{d\Omega}\right|_{\text{meas}}}{\left.\frac{d\sigma}{d\Omega}\right|_{\substack{\text{point like}\\ \text{target}}}} \equiv |F(q^2)|^2$$

$f(\vec{r}) = f(r)$

$$F(q^2) = \int d^3r \; f(\vec{r}) \; e^{+i\vec{q}\vec{r}} = \int_0^{2\pi} d\varphi \int_{-1}^{1} d\cos\theta' \int_0^{\infty} dr \, r^2 \; (\,-\,-\,)$$

$$e^{+i\vec{q}\cdot\vec{r}} \; f(r)$$

$$Q^2 = -q^2 = 4EE' \sin^2\frac{\theta}{2}$$

probing small q^2 easier

Look at the limit $q^2 \to 0$

$$e^{i\vec{q}\cdot\vec{r}} = e^{iqr\cos\theta'} \simeq 1 + i\,qr\cos\theta' - \frac{1}{2}(qr)^2\cos^2\theta' + O(q^3)$$

$$\int_{-1}^{1} d\cos\theta' \left(\underbrace{1}_{2} + \underbrace{i\,qr\cos\theta'}_{\emptyset} - \underbrace{\frac{1}{2}(qr)^2\cos^2\theta'}_{-\frac{1}{6}(qr)^2} \right) =$$

for small q^2

$$F(q^2) \simeq 4\pi \underbrace{\int_0^\infty dr\, f(r)\, r^2}_{A} + 0 - \frac{1}{6}\, 4\pi\, q^2 \int_0^\infty dr\, f(r)\, r^4$$

$$F(q^2) = 1 - \frac{1}{6}\, q^2 \langle r^2 \rangle \qquad \langle r^2 \rangle = \int d^3r\, f(r)\, r^2$$

$$\rho(r) = Z_T e\, f(r) \qquad \int \rho(r')\, d^3r' = Z_T$$

$$\int d^3r\, Z_T e\, f(r) = Z_T e$$

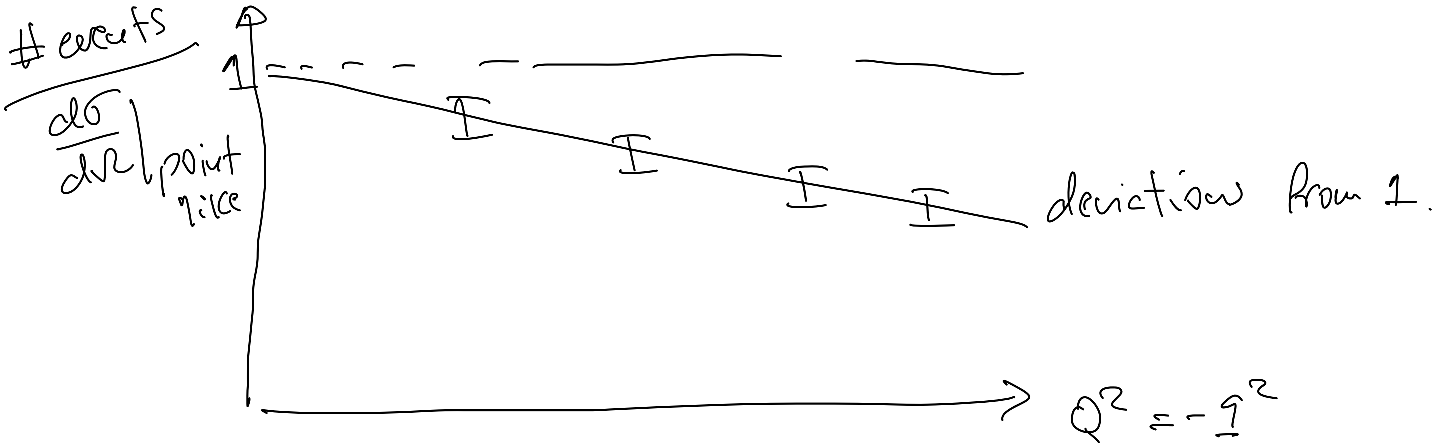
$$\Rightarrow \int d^3r\, f(r) = 1$$

$$F(q^2) \simeq 1 - \frac{1}{6} q^2 \underbrace{\langle r^2 \rangle}_{\geq 0} \qquad \int d^3r\, r^2 f(r)$$

≤ 1 for $q^2 \geq 0$ small q^2

$$\frac{dF(q^2)}{dq^2} = -\frac{1}{6} \langle r^2 \rangle$$

$q^2 \to 0 \quad F(q^2) = 1$



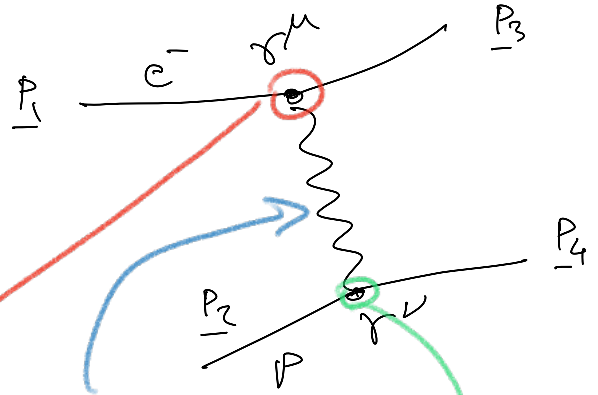
$F(q^2) \neq 1 \Rightarrow \langle r^2 \rangle \neq 0 \Rightarrow$ target not point like.

Dirac proton target

spin = 1/2 e^- prob

spin = 1/2 proton target

point like proton.



$$\mathcal{M} \sim -ie\left[\bar{u}(p_3)\gamma^\mu u(p_1)\right] \frac{-ig^{\mu\nu}}{q^2} \left[\bar{u}(p_4)\left(\gamma^\nu + \frac{i}{2}\frac{\sigma^{\nu\tau}q_\tau}{M}\kappa\right)u(p_2)\right]$$

$\kappa = g-1$

spin for proton $\Rightarrow$ take into account magnetic moment

point-like particle $\quad \mu = g\dfrac{e\hbar}{4m} \quad$ g: gyromagnetic factor

e^-: $g = 2 \quad \mu_e = \dfrac{e}{2m} \quad \hbar = 1$

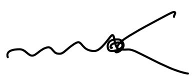
p: $g = 2$ pointlike. $\quad g_p = 2.79$ measured.

n: $g = \emptyset$ pointlike expected $\quad g_n = -1.91$ measured

γ^μ: Dirac matrices.

$$\gamma^\nu + \frac{i}{2}\frac{\sigma^{\nu\tau}q_\tau}{2M}\kappa \qquad \sigma^{\nu\tau} = \frac{i}{2}\left[\gamma^\nu, \gamma^\tau\right]$$

$\kappa = g-1$



Dirac proton vertex: $s = 1/2$ pointlike.

Rosenbluth Formula

$$\frac{d\sigma}{d\Omega} = \frac{(\alpha Z_p Z_T)^2}{q^4} E'^2 \underbrace{\left(1-\beta^2\sin^2\frac{\vartheta}{2}\right)}_{\cos^2\frac{\vartheta}{2}} \frac{E'}{E}\underbrace{\left(1 + \frac{Q^2}{2M^2}\tan^2\frac{\vartheta}{2}\right)}_{\text{Dirac proton}}$$

$S = 1/2$. $\mu \neq 0$

$\beta \simeq 1$

$$\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega}\right)_{\text{Ruth.}} \underbrace{\frac{E'}{E}}_{\text{Recoil}} \left(\cos^2\frac{\vartheta}{2} + \frac{Q^2}{2M^2}\sin^2\frac{\vartheta}{2}\right)$$

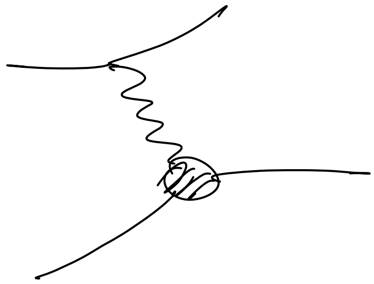
$\dfrac{Q^2}{2M^2}$ For $Q^2 \ll (M_{proton})^2$ $M_p^2 \simeq 1\ \text{GeV}^2$.

$q^2 = 4EE' \sin^2\frac{\theta}{2}$ for $E < 1$ GeV.
difficult to notice
Dirac Proton.

<u>Real experiment</u>

e^- $S = 1/2$ $\beta \simeq 1$ $E_e > 100$ MeV

proton $S = 1/2$ assume Charge distribution (structure)



$$\underbrace{F_1(q^2)\gamma^\nu}_{\text{modification of charge interaction}} + i\frac{\sigma^{\nu\tau}q_\tau}{2M}\underbrace{K\ F_2(q^2)}_{\text{magnetic mom of structure}}$$

non-pointlike.

Rosenbluth + Form Factors

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{non-pointlike proton.}} = \frac{(\alpha z_e z_p)^2}{q^4} E'^2 \frac{E'}{E}\left[\left(F_1^2 + \frac{K^2Q^2}{4M^2}F_2^2\right)\cos^2\frac{\theta}{2} + \right.$$

$$\left.(F_1 + KF_2)^2\frac{Q^2}{2M^2}\sin^2\frac{\theta}{2}\right]$$

$$M \sim \overset{\text{electric}}{(F_1\ )} + \overset{\text{magnetic part}}{(--\ F_2)} = \bar{u}(p_4)(\gamma^\nu F_1 + --)u(p_2)$$

$$|M|^2 = M^\dagger M = (a^\dagger + b^\dagger)(a+b) = a^\dagger a + b^\dagger b + a^\dagger b + b^\dagger a$$

$Q^2 \ll M^2$ only depends on $|F_1(q^2)|^2$

$$\lim_{Q^2\to 0} |F_1(q^2)|^2 = 1$$

$$\frac{Q^2}{M^2} \ll 1 \rightarrow \frac{d\sigma}{d\Omega} \simeq \frac{(\alpha' Z_T Z_p)^2}{q^4} E'^2 \frac{E'}{E} \cos^2\frac{\theta}{2} \, |F_1(Q^2)|^2$$

small $Q^2 \Rightarrow$ dominated by F_1.

$$\frac{Q^2}{M^2} \gg 1 \quad \frac{d\sigma}{d\Omega} \simeq \frac{(\alpha Z_T Z_p)^2}{q^4} E'^2 \frac{E'}{E} \times \left(\kappa^2 |F_2(q^2)|^2 \frac{Q^4}{4M^4} \right)$$

$$Q^2 = 4EE' \sin^2\frac{\theta}{2}$$

Form factors for proton

McAllister, Hofstadter (1956) at SLAC

Nobel 1961

beam of e^- $E = 188$ MeV

$e^- + p \rightarrow e^- + p$

$e^- + \alpha \rightarrow e^- + \alpha$

Goldhaber–Cahn Chap. 8

$\theta \in [35°, 138°]$

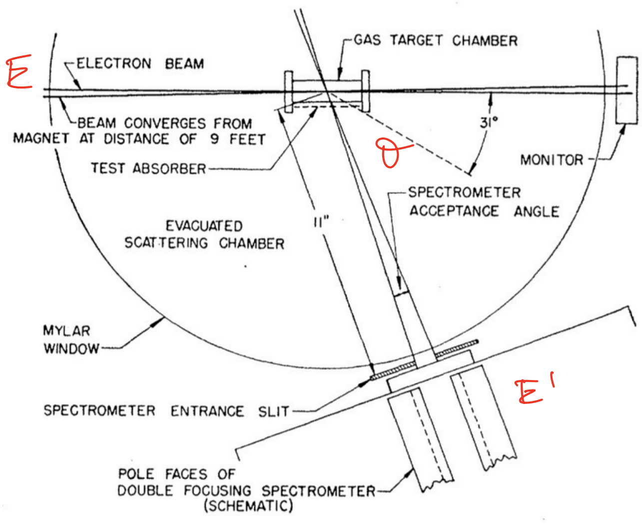
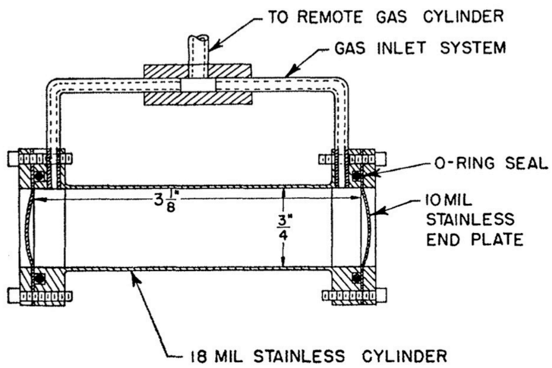
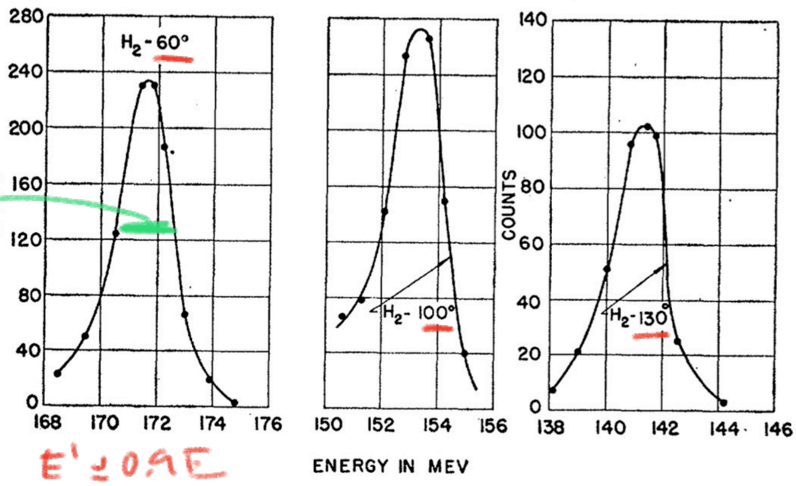


FIG. 2. Arrangement of parts in experiments on electron scattering from a gas target.



$$E' = \frac{E}{1 + \frac{E}{M}(1 - \cos\theta)}$$

Detector resolution



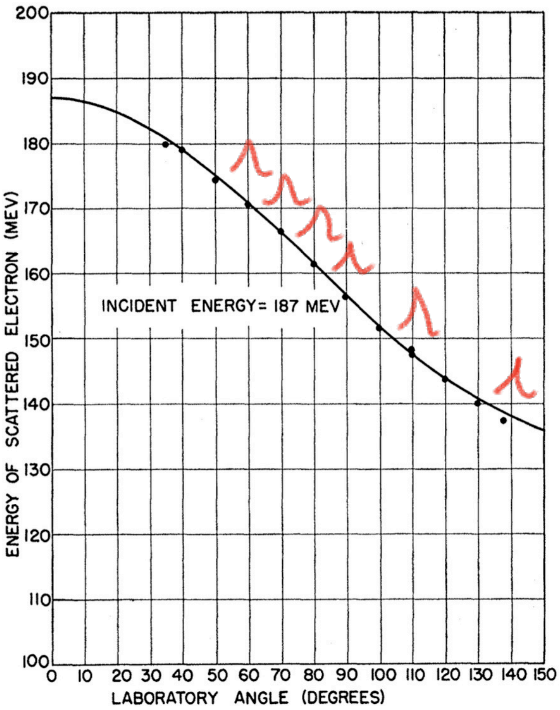
$E' \simeq 0.9E$

$60°: \cos\theta = \frac{1}{2}$

$$E' = \frac{1}{1 + \frac{E}{M}\frac{1}{2}}$$

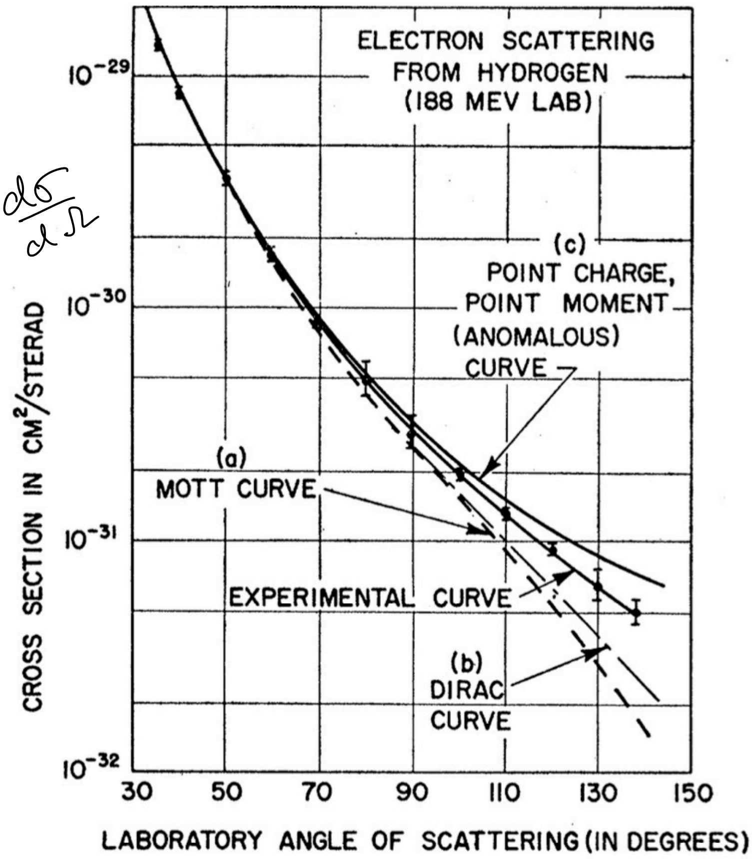
$$\frac{E}{M} = \frac{0.2}{1}$$

$$\frac{E'}{E} = \frac{1}{1+0.2\times\frac{1}{2}} = \frac{1}{1+0.1} \simeq 1-0.1 = 0.9$$



$$E' = \frac{E}{1+\frac{E}{M}(1-\cos\theta)}$$

$\frac{d\sigma}{d\Omega}$



relativistic electron.

a) proton - point like
no spin
no structure
} Mott.

b) no structure $F_1 = F_2 = \emptyset$
$g = 2$

c) $g \neq 2$
no structure.

$$\left.\frac{d\sigma}{d\Omega}\right|_{meas} = \left.\frac{d\sigma}{d\Omega}\right|_{point} \times |F(q^2)|^2$$

$$F(q^2) = 1 - \frac{1}{6} q^2 \langle r^2 \rangle$$

$$Q^2 = 4EE' \sin^2\frac{\theta}{2}$$

$|F(q^2)|^2 \neq 1 \Rightarrow$ estimate $\langle r^2 \rangle$ from data

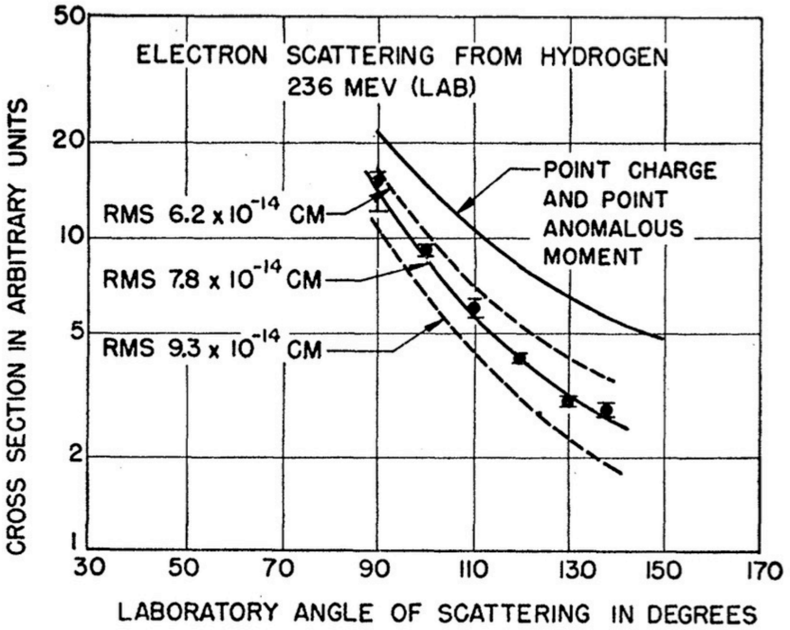


FIG. 6. This figure shows the experimental points at 236 Mev and the attempts to fit the shape of the experimental curve. The best fit lies near 0.78×10^{-13} cm.

$\mathrm{RMS} = \sqrt{\langle r^2\rangle}$

$\sqrt{\langle r^2\rangle} = 7.8\times10^{-14}$ cm

$\Delta\chi^2 = 1 \Rightarrow$ estimate of uncertainty

$\sqrt{\langle r^2\rangle_p} \simeq 0.78$ fm

$r_N = r_0 A^{1/3}$ $\qquad r_0 = 1.1$ fm

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Elastic Scattering of 188-Mev Electrons from the Proton and the Alpha Particle*†‡§‖¶

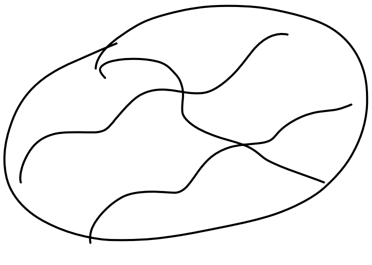
R. W. McALLISTER AND R. HOFSTADTER
Department of Physics and High-Energy Physics Laboratory, Stanford University, Stanford, California

(Received January 25, 1956)

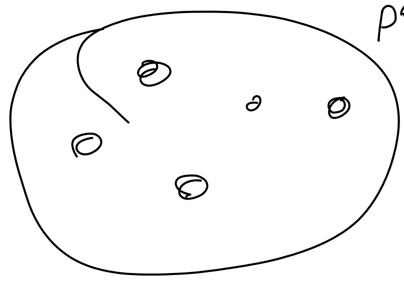
The elastic scattering of 188-Mev electrons from gaseous targets of hydrogen and helium has been studied. Elastic profiles have been obtained at laboratory angles between 35° and 138°. The areas under such curves, within energy limits of ±1.5 Mev of the peak, have been measured and the results plotted against angle. In the case of hydrogen, a comparison has been made with the theoretical predictions of the Mott formula for elastic scattering and also with a modified Mott formula (due to Rosenbluth) taking into account both the anomalous magnetic moment of the proton and a finite size effect. The comparison shows that a finite size of the proton will account for the results and the present experiment fixes this size. The root-mean-square radii of charge and magnetic moment are each $(0.74\pm0.24)\times10^{-13}$ cm. In obtaining these results it is assumed that the usual laws of electromagnetic interaction and the Coulomb law are valid at distances less than 10^{-13} cm and that the charge and moment radii are equal. In helium, large effects of the finite size of the alpha-particle are observed and the rms radius of the alpha particle is found to be $(1.6\pm0.1)\times10^{-13}$ cm.

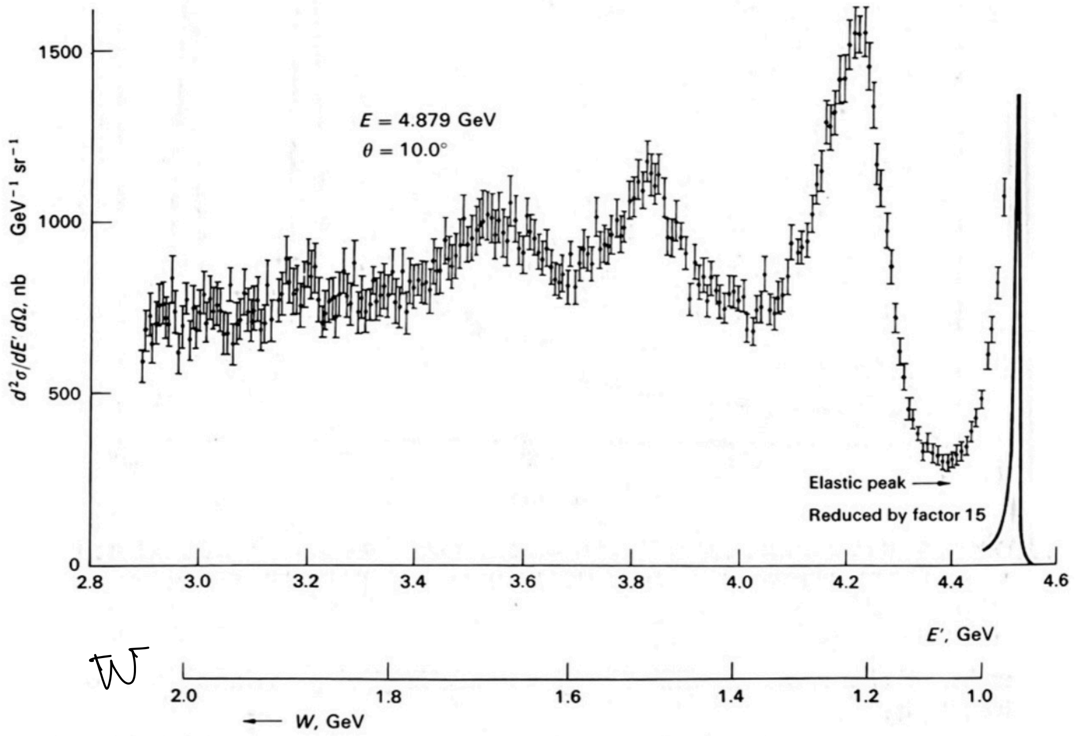
$\Rightarrow$ proton has structure

continuous



aggregation of point like objects.



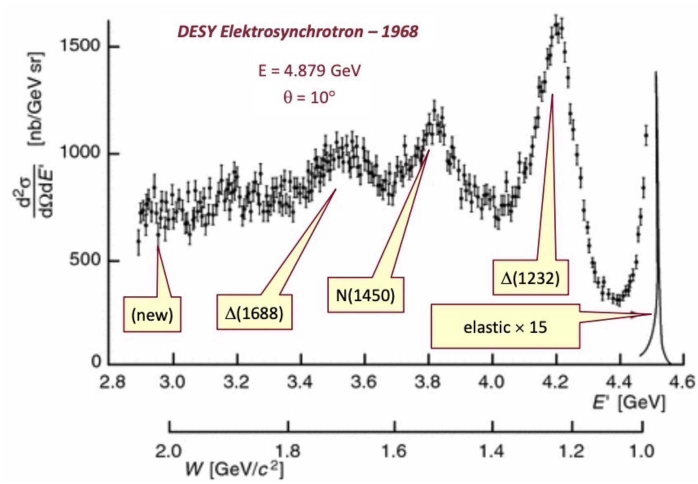


$W = \sqrt{P_H^2}$

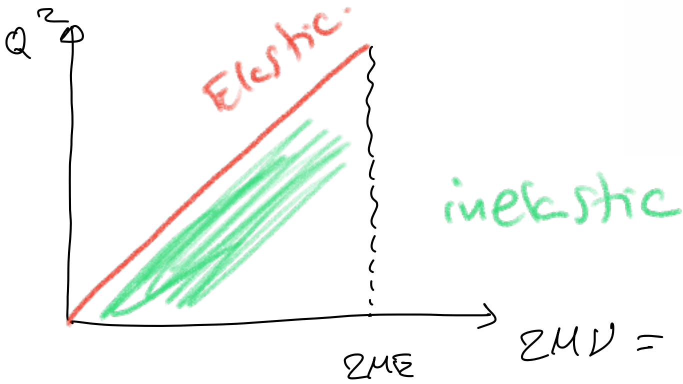
Figure 8.6 Excitation curve of inelastic *ep* scattering, obtained at the DESY electron accelerator (Bartel *et al.* 1968). *E* and *E'* are the energies of the incident and the scattered electron, and *W* is the mass of the recoiling hadronic state. The peaks due to the pion-nucleon resonances of masses 1.24, 1.51, and 1.69 GeV are clearly visible.

$e^- + p \longrightarrow e^- + p$

$e^- + X$



$Q^2 = M^2 - W^2 + 2M(E - E')$



$2M\nu = 2M(E - E')$

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{inelastic}} (e^- + p \to e^- + X) = \frac{\alpha^2}{q^4} E'^2 \cos^2\frac{\theta}{2} \left(W_2(Q^2,\nu) + 2W_1(Q^2,\nu) \tan^2\frac{\theta}{2} \right)$$

W_1, W_2 : structure functions

expressing our ignorance about the proton structure

$E = 5$ GeV $\quad \frac{Q^2}{M^2} \geq 1$

Experimentally, measure $E', \theta \Rightarrow Q^2, \nu$

count # events as function of Q^2, ν

Fit the data with functional form including W_1, W_2 functions.

$$x = \frac{Q^2}{2M\nu} \qquad \in [0,1]$$

$$y = \frac{\nu}{E} = \frac{E-E'}{E} \qquad \in [0,1]$$